

Energy-Efficient and Dequantization-Free Quantization of LLMs: A Spiking Neural Network Approach to Salient Value Mitigation

Chenyu Wang
Sun Yat-sen University
Guangzhou, Guangdong, China
wangcy236@mail2.sysu.edu.cn

Zhanglu Yan*
National University of Singapore
Singapore
zlyan@nus.edu.sg

Zhi Zhou*
Sun Yat-sen University
Guangzhou, Guangdong, China
zhouzhi9@mail.sysu.edu.cn

Xu Chen
Sun Yat-sen University
Guangzhou, Guangdong, China
chenxu35@mail.sysu.edu.cn

Weng-Fai Wong
National University of Singapore
Singapore
wongwfw@nus.edu.sg

Abstract

In the era of large language models (LLMs), weight-activation quantization helps fit models on edge device by reducing memory and compute bit-widths. However, three challenges persist for energy constrained hardware: (1) even after quantization, multiply-accumulate (MAC) operations remain unavoidable and continue to dominate energy consumption; (2) dequantization (or per-tensor/channel rescaling) introduces extra arithmetic and data movement, increasing latency and energy; (3) uniform parameters bit widths clip salient values—while intra-channel mixed precision is generally impractical on current matrix hardware and memory. In contrast, brain-inspired Spiking Neural Networks (SNNs), owing to their binary spike-based information representation and the Integrate-and-Fire (IF) paradigm, naturally support mixed-precision storage and energy-efficient computation by replacing complex MACs with temporal Accumulate (ACCs). Motivated by this property, we propose SpikeQuant, which selectively applies mixed-precision quantization to activations with salient values and re-encodes them into binary spike counts, thereby enabling dynamic mixed storage of different bitwidths. Furthermore, by embedding the quantization scale into the threshold of the IF mechanism, our approach performs energy-efficient linear transformations on weights and activations while avoiding explicit dequantization. Experimental results demonstrate that SpikeQuant consistently achieves near-FP16 perplexity under W4A4 quantization while reducing energy cost by up to 4.6× compared to existing methods, highlighting its effectiveness for accurate and energy-efficient LLM deployment. Our code is open-sourced at <https://github.com/wangchenyu929/SpikeQuant.git>

CCS Concepts

• Computing methodologies → Neural networks.

*Corresponding authors are Zhi Zhou and Zhanglu Yan.



This work is licensed under a Creative Commons Attribution 4.0 International License. *WWW '26, Dubai, United Arab Emirates.*
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2307-0/2026/04
<https://doi.org/10.1145/3774904.3792337>

Keywords

Large Language Model, Weight Activation Quantization, Silent Value Mitigation, Spiking Neural Networks, Energy Efficiency Inference

ACM Reference Format:

Chenyu Wang, Zhanglu Yan, Zhi Zhou, Xu Chen, and Weng-Fai Wong. 2026. Energy-Efficient and Dequantization-Free Quantization of LLMs: A Spiking Neural Network Approach to Salient Value Mitigation. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3774904.3792337>

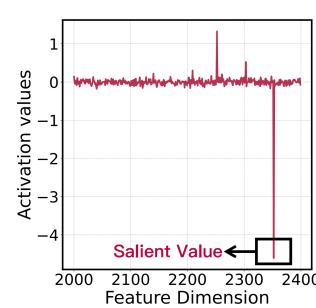


Figure 1: Activations distribution of Llama2-7B during inference

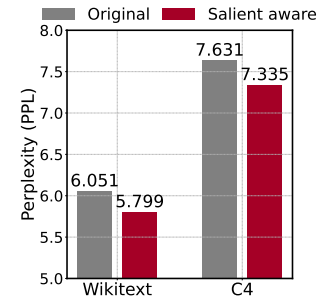


Figure 2: Perplexity of original and salient aware quantization of Llama2-7B

1 Introduction

In recent years, Artificial Neural Networks (ANN) based LLMs with billions of parameters [1] [2] have greatly advanced applications in browser inference engines [3] and web enhanced question-answering systems [4]. Concurrently, there has been a growing demand for deploying such models in Web-of-Things (WoT) environments, driven by privacy, latency considerations for devices in mobile WoT settings. However, memory and energy budgets remain the bottleneck—for instance, FP32 Llama2-7B alone stores 28 GB of weights and a single NVIDIA A6000 draws 72 W over 10 s of inference [5]. To address these issues, weight-activation quantization converts full-precision parameters into low-bit representations (i.e., INT8 or INT4), thereby reducing the memory footprint and accelerating inference [6].

However, accurate quantization LLMs are hard: activations frequently contain salient outliers that, under low bit-widths, set an overly wide scale, compressing typical values and hurting accuracy, as shown in Fig. 1 and Fig. 2. Such issue is typically more severe than for weights, whose post-training distributions are near-Gaussian and relatively stable [7, 8]. Atom[9] and AWQ [10] mitigate salient values via salient channel reordering or per-channel rescaling. However, they rely on time consuming offline calibration and search processes, and their fine-grained scaling strategies (e.g., intra-channel or group-wise) are expensive on current hardware, leading to tensor fragmentation, complicated scheduling and higher energy consumption.

Compared to ANN-based LLMs, the human brain, with roughly 86 billion neurons, achieves far greater efficiency. Based on this inspiration, Spiking Neural Networks (SNNs) [11], which leverage binary spike-based information and the Integrate-and-Fire (IF) paradigm, have emerged as a promising approach for next-generation machine intelligence. The advantage of SNNs lies in the full utilization of the temporal dimension, where parameters are transmitted as 0/1 spikes across T steps, and the conversion between different quantization bits only requires changing T . When applied to the quantization of ANNs, this allows for (1) performing in-place mixed quantization of salient and normal values without channel reorder or channel rescaling, and (2) replacing Multiply-Accumulate (MAC) operations with energy-efficient spike Accumulations (Acc), which reduces the computation energy [12].

Despite their advantages for quantizing salient values, the application of SNNs remains limited to smaller models (e.g., VGG, ResNet, and BERT) [13–16]. This limitation primarily stems from the substantial memory and data movement costs of rate encoding (where a parameter’s value corresponds to its spike count over T time steps) and the information loss incurred when the residual membrane potential is discarded in standard IF neurons from improper thresholding.

In terms of parameter representation, we address the salient value phenomenon—which is more pronounced in activations and more challenging to handle due to their real-time generation during inference—by encoding activations into a time-to-first-spike (TTFS) scheme [17], where information is represented solely by the latency of the first spike. Compared with the rate encoding commonly used in SNNs, TTFS requires only a single spike within a time window, thereby significantly reducing data movement costs and memory footprint. Regarding the IF mechanism, we define the firing threshold as a function of the scales in mixed-precision quantization, such that the same mathematical effect as dequantization can be achieved without explicitly performing the dequantization process. As far as we know, in the ANN domain, this is the first work to leverage TTFS-based SNNs for mitigating salient values, improving model accuracy while reducing inference energy consumption; in the SNN domain, we bridge the gap by introducing a scale-based threshold, enabling SNN-LLMs to achieve accuracy comparable to, or even on par with, their ANN counterparts.

Our main contributions are summarized as follows:

- We propose SpikeQuant, the first framework that leverages TTFS encoding in quantized LLMs, effectively mitigates the impact of salient values.

- We embed mixed-precision quantization scales into the fire threshold in SNN, ensuring mathematical equivalence to dequantized linear transformations without explicitly performing dequantization.
- We conduct extensive motivation studies and experiments, demonstrating that SpikeQuant achieves near-FP16 perplexity and up to 78.5% energy savings compared to state-of-the-art baselines, and narrows the performance gap between SNN-LLMs and ANN-LLMs.

2 Preliminaries

2.1 Low-bit Quantization

Low-Bit quantization maps a real-valued tensor x to an integer tensor x_{int} via a scale $S > 0$ and a zero-point $Z \in \mathbb{Z}$. The asymmetric quantization and dequantization steps are typically defined as:

$$x_{\text{int}} = \text{clamp}\left(\text{round}\left(\frac{x}{S}\right) + Z, q_{\min}, q_{\max}\right), \quad (1)$$

$$\hat{x} = S(x_{\text{int}} - Z), \quad (2)$$

where $q_{\min}, q_{\max}$ denote the limits of the integer range (e.g. for unsigned 8-bit, $[0, 2^8 - 1]$), and clamp ensures values stay in that range. In contrast, symmetric quantization sets the zero-point $Z = 0$, and chooses the scale based on the maximum of the absolute value

$$S = \frac{\max(|r_{\min}|, |r_{\max}|)}{q_{\max}}, \quad (3)$$

where $[r_{\min}, r_{\max}]$ is the clipping range of the floating values.

Asymmetric quantization, by incorporating a zero-point offset, more accurately represents skewed or non-symmetric data distributions, thereby minimizing quantization error compared to the rigid, zero-centered range of symmetric methods. Therefore, the quantization method selected in this paper is asymmetric quantization

2.2 Spiking Neural Networks

We firstly focus on the IF paradigm in SNN: a neuron maintains a membrane potential $V(t)$ that integrates incoming inputs over discrete time steps. We denote $I(t)$ as the synaptic input at time t , and the neuron emits a spike $s(t)$ if $V(t-1) + I(t)$ exceeds or equals a threshold V_{th} , and otherwise remains silent:

$$V(t) = V(t-1) + I(t) - s(t) \cdot V_{\text{th}}, \quad (4)$$

$$s(t) = \begin{cases} 0, & V(t-1) + I(t) < V_{\text{th}}, \\ 1, & V(t-1) + I(t) \geq V_{\text{th}}. \end{cases} \quad (5)$$

There are several ways to encode input or activation values using spike trains [18, 19]. The most common is rate encoding, where information is represented by the number (or frequency) of spikes over a time window T . Although simple, this incurs a large number of spikes, increasing data movement, and memory cost. A more efficient alternative is TTFS encoding, where each neuron is allowed at most one spike per window T . The timing, or latency, of that first spike encodes the magnitude (stronger or more urgent inputs spike earlier). TTFS reduces the total number of spikes and can dramatically lower memory access overhead compared to rate encoding.

3 Motivation Experiments

This section examines the influence of quantization bitwidth on both activations and weights, as well as the distribution of activations during inference. Building on these observations, we further discuss the challenges that arise when implementing mixed-precision quantization.

3.1 Importance of Activations

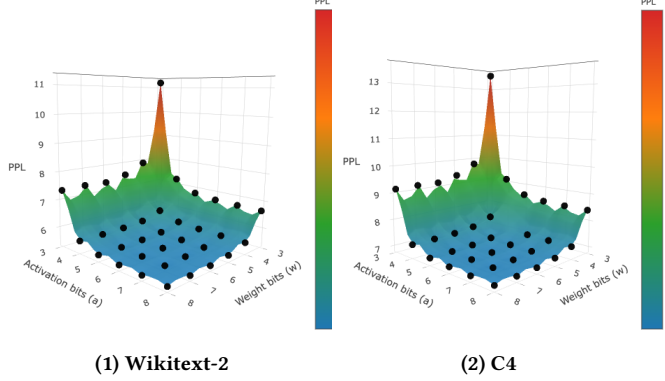


Figure 3: Perplexity of Llama-2-7B across different weight-activation quantization bitwidths

Weight-activation quantization has emerged as a prominent strategy for efficient model compression[20]. To investigate the respective impacts of quantization bitwidth on weights and activations, we evaluate Llama2-7B under 1–8 bit quantization on the Wikitext-2 [21] and C4 [22] datasets, with perplexity (PPL) as the metric. The results are presented in Fig. 3.

Under extremely low-bit quantization settings (e.g., 1-bit and 2-bit), LLMs suffer from severe information loss, leading to a collapse in performance, as indicated by perplexity values rising to the order of 10^5 . Therefore, we report only the results for 3–8 bit quantization. As shown in Fig. 3, increasing activation bitwidth reduces perplexity more effectively than increasing weight bitwidth, with the 4w4a configuration offering a strong balance between efficiency and performance. However, further bitwidth increases yield diminishing returns, limiting the optimal trade-off between memory footprint and accuracy.

Based on the above experimental conclusions and the importance of activations, we quantize both weights and activations to 4-bit, while quantizing the activation salient values to 5-bit. Unlike prior works that typically use 8-bit or higher for salient values [9, 23], our approach avoids the unjustified memory overhead often associated with such high-precision schemes.

3.2 Distribution of Activations

In weight-activation quantization, activation salient values (extreme magnitudes) are a primary cause of accuracy degradation in LLMs [24]. To characterize this phenomenon more precisely, we analyze the activation statistics of Llama2-7B across all 32 transformer layers and seven representative linear modules ($gate_{proj}$, $down_{proj}$, up_{proj} , k_{proj} , q_{proj} , v_{proj} , o_{proj}) during a single forward pass with the input token “is”. Each activation tensor has shape

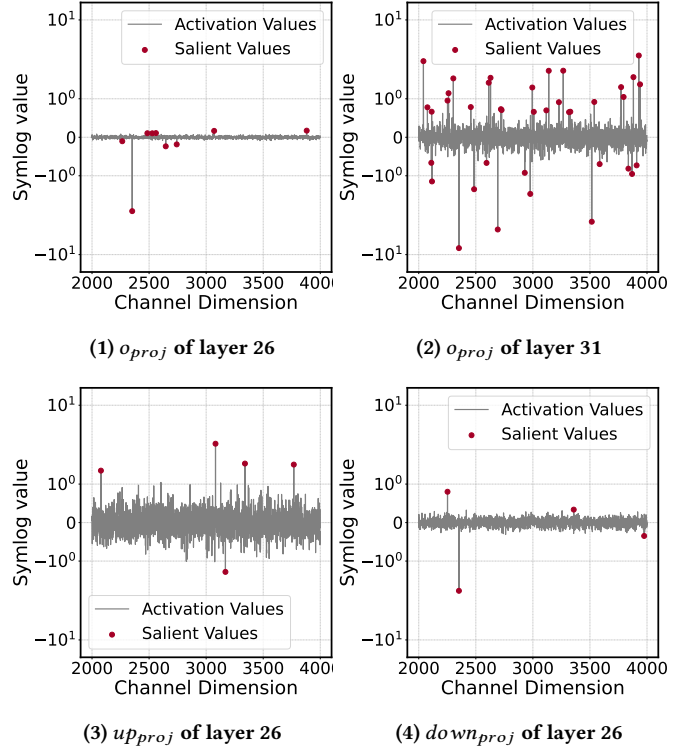


Figure 4: Activation distributions and salient values in Llama-3-8B under different module-layer combinations

$[B, S, H]$, where $B=1$, $S=1$, and $H=4096$. Salient activations, detected via the Median Absolute Deviation (MAD) algorithm (detailed in Section 4.1), are highlighted in red, with the complete layer-wise visualization provided in Appendix C.2.

Figures 4-1 to 4-4 illustrate the heterogeneity of activation distributions across layers and modules. The results reveal that salient activations are neither uniformly distributed nor temporally consistent. For example, even within the same module (o_{proj}), different layers exhibit distinct tail behaviors and outlier densities, indicating strong inter-layer variation in dynamic ranges. Likewise, within the same layer, the q_{proj} , v_{proj} , and up_{proj} modules produce markedly different activation kurtosis and sparsity levels, reflecting module-specific quantization sensitivity. These observations highlight that activation statistics are inherently non-stationary and token-dependent, which complicates static quantization schemes that assume fixed or channel-wise invariant distributions.

3.3 Challenges

To address activation salient values in model quantization, a common strategy is to assign higher bitwidths to salient values through mixed-precision quantization [10, 25]. Despite its advantages in model compression, mixed-precision quantization during inference faces three key challenges:

1) Dynamic distribution of activation salient values. The distribution of salient values varies across layers, modules, and even individual tokens. Consequently, fixed-channel strategies fail to

adequately capture all salient values, thereby limiting the accuracy of mixed-precision quantization and degrading model performance.

2) Incompatibility of storage between salient and normal values. Once activation salient values are quantized to high-bit, they cannot be stored together with low-bit normal values. This necessitates simultaneous reordering of both weight and activation channels, which incurs significant computational overhead.

3) Increased overhead in linear transformations. During linear operations, both activations and weights require costly dequantization procedures. This results in increased peak memory footprint and energy consumption, which may ultimately hinder or stall inference.

4 SNN-based LLM Inference Framework Design

To better handle salient values in ANN-based LLM quantization and address the above three challenges, we propose SpikeQuant. Specifically, we (i) use the offline MAD algorithm to obtain the salient bar for online accurate but low-cost detection of dynamic salient value, (ii) leverage TTFS encoding to convert quantized activations into binary spikes for mixed-precision storage, and (iii) integrate mixed-precision quantization scales into the SNN firing threshold, thereby avoiding both dequantization and costly dense MACs.

4.1 MAD-Guided Salient Activation Detection

Different from prior works [26, 27], we adopt the Median Absolute Deviation (MAD) [28], a robust statistical estimator for outlier detection, to identify salient values along the hidden dimensions of each module’s activations during LLM inference.

The use of the MAD offers two key benefits. First, MAD is insensitive to extreme values, since the median (and deviations from it) is not strongly affected by outliers, unlike the standard deviation [26] which can be inflated by them. Second, unlike top-k methods [27] that rely on selecting a fixed number or fixed proportion of “largest” values, MAD-based thresholds adapt to the actual data distribution: when the number of salient values is unstable (varying across layers or inputs), a MAD-based criterion will still identify outliers accurately without the need to tune how many to pick.

For activations x , MAD is defined as

$$\text{MAD} = \text{median}(|x_i - \text{median}(x)|), \quad (6)$$

and the standardized score is computed as

$$M_i = \frac{x_i - \text{median}(x)}{c \cdot \text{MAD}}, \quad (7)$$

where c is a scaling constant (typically $c = 1.4826$ [29]) ensuring that the expected value of $c \cdot \text{MAD}$ to be roughly equal to the standard deviation for normally distributed data. Given a MAD threshold r (commonly $r = 3.5$ [29]), any activation with $|M_i| > r$ is regarded as a salient value.

To avoid additional computational overhead during inference, we follow prior work [9, 30] and employ 128 randomly sampled sentences from the WikiText2 dataset as calibration data for offline salient value detection. Concretely, we store the average of the positive salient minimum and the negative salient maximum across 128 forward passes, referred to as the salient bar, which is then used as the criterion for online detection. This approach not only mitigates

the limitations of fixed salient channels—which cannot capture all potential salient values—but also circumvents the expensive sorting operations required by top- k selection methods. Overall, it enables a lightweight yet accurate mechanism for dynamically identifying salient values during inference.

4.2 TTFS Encoding of Mixed-Precision Activations

After identifying salient values, we quantize them with a higher bitwidth (i.e. 5-bit) to preserve precision, while the majority of normal values are quantized with a lower bitwidth (i.e. 4-bit) using group-wise scaling [27, 31], thus better adapting the quantized range to local activation distributions. Specifically, in a single linear transformation module, activations have shape $[B, S, H]$ (batch, sequence, hidden) and weights have shape $[In, Out]$, producing outputs $[B, S, Out]$. Quantization groups (G groups) are formed along the hidden dimension H for activations and the input dimension In for weights, while salient activation detection is performed along H , as illustrated in the left part of Fig. 5

In practical hardware, however, mixed-bit activations cannot be directly stored in a single tensor due to constraints such as memory alignment requirements, SIMD/Systolic-array execution granularity, and the lack of heterogeneous-precision support in tensor cores [32]. To resolve this limitation, we transform quantized activations into TTFS spike trains in the SNN domain.

Formally, given a quantized activation value $Qa \in \{0, 1, \dots, 2^b - 1\}$ with bitwidth b , we define the encoding window length as $T = 2^b - 1$. Under TTFS encoding, the corresponding spike train is

$$a(t) = \begin{cases} 1 & \text{if } t = T - Qa \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

Eq. (8) means that the larger the Qa value, the higher the membrane potential strength, and the earlier the spike occurs, as shown in Fig. 5. This binary encoding method provides a different time window mapping across mixed precisions without requiring heterogeneous storage.

From a hardware perspective, spike generation is executed in parallel across activation channels. Each channel is equipped with a local spike generator, typically implemented using counters or stochastic firing units, which emits spike trains according to the pre-computed firing rate. Because all activations are encoded within the same wall-clock window T , parallel spike streams are temporally aligned for subsequent synaptic operations. During computation, spikes serve as binary events that are multiplied with synaptic weights. This spike-weight multiplication is efficiently realized in neuromorphic accelerators via event-driven accumulate-and-shift operations, where each spike triggers the addition of the corresponding weight to the postsynaptic membrane potential. In this way, mixed-precision activations are seamlessly integrated into the SNN computation pipeline without requiring direct mixed-bit tensor storage.

4.3 Dequantized ANN Equivalent SNN Inference

After establishing the binary TTFS spike-based representation of activations, our objective is to ensure that the outputs of SNNs under mixed-precision quantization remain consistent with those

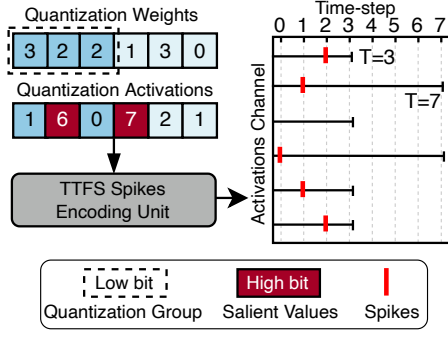


Figure 5: Group weight-activation quantization and TTFS encoding

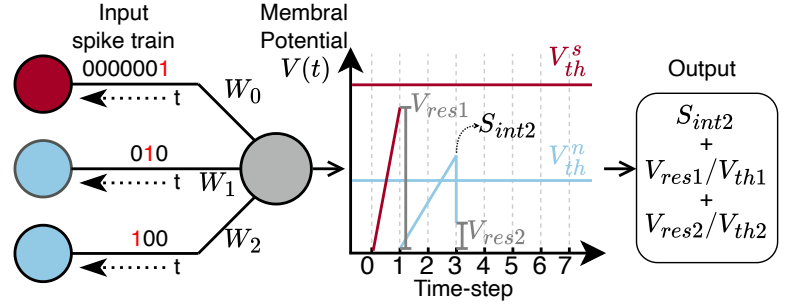


Figure 6: Convertible SNN threshold: membrane potential accumulation and firing process

obtained from the dequantized linear projections in ANN-based LLMs. Specifically, within a quantization group that contains only low-bit normal values, which contains no high-bit salient values, the dequantized linear transformation in a quantization group of the input sequence is expressed as

$$\begin{aligned} Output_{ANN} &= \sum_{i=1}^{H_g} (Qa_i - Za) \cdot Sa \cdot (Qw_i - Zw) \cdot Sw \\ &= \underbrace{Sa \cdot Sw}_{z1} \cdot \underbrace{\sum_{i=1}^{H_g} [Qa_i \cdot (Qw_i - Zw) + Za \cdot (Zw - Qw_i)]}_{z2} \end{aligned} \quad (9)$$

where H_g denotes the quantization group size, Qa_i and Qw_i are the quantized activations and weights, respectively. (Sa, Za) and (Sw, Zw) represent the scaling factors and zero-points for activations and weights. Under symmetric quantization, the zero-points vanish.

From a mathematical perspective, the IF mechanism in SNNs can be interpreted as an accumulation of membrane potential V_{acc} , followed by emitting spikes whenever V_{acc} exceeds the firing threshold V_{th} . In particular, the number of emitted spikes S_{int} corresponds to the integer quotient of the membrane potential and the threshold:

$$S_{int} \triangleq \left\lfloor V_{acc} \cdot \frac{1}{V_{th}} \right\rfloor. \quad (10)$$

Based on Eq.(9), we observe that if we consider Qa_i as TTFS spike train and $(Qw_i - Zw)$ as potential weight, the $z2$ term can be regarded as the accumulated potential V_{acc} in SNNs:

$$V_{acc} = \underbrace{\sum_{i=1}^{H_g} (T - t_i) \cdot (Qw_i - Zw)}_{\text{membrane potential } V} + \underbrace{\sum_{i=1}^{H_g} Za \cdot (Zw - Qw_i)}_{\text{SNN bias } b}, \quad (11)$$

where t_i denotes the spike arrival time of element i , and T is the encoding time window length of activation Qa_i . Based on $z1$ in Eq.(9), Eq. (10) and Eq. (11), if we take $1/(Sa \cdot Sw)$ as V_{th} , then the fire spike count equals the integer part of the ANN output, which we denoted as S_{int} . The SNNs IF process is

$$V(t) = V(t-1) + a(t) \cdot (T-t) \cdot (Qw_i - Zw) + b(t) - S(t)V_{th}. \quad (12)$$

$$S(t) = \begin{cases} 0, & \text{if } V(t) < V_{th} \\ 1, & \text{if } V(t) \geq V_{th} \end{cases}. \quad (13)$$

$$b(t) = \begin{cases} \sum_{i=1}^{H_g} Za \cdot (Zw - Qw_i), & \text{if } t = T \\ 0, & \text{otherwise} \end{cases}. \quad (14)$$

Eq. (12) shows that the membrane potential at time t accumulates both the previous potential $V(t-1)$ and the contribution of the new spike. In TTFS coding, $a(t) \cdot (T-t)$ indicates that a spike at time t corresponds to $(T-t)$ equivalent spikes, while $a(t) = 0$ otherwise. The term $(Qw_i - Zw)$ represents the spike weight, and $-S(t)V_{th}$ with Eq. (13) ensures that once the potential exceeds the threshold, a spike is emitted and $V(t)$ decreases by V_{th} . Eq. (14) further adds a bias term to the residual potential for a final firing. This bias $b(t)$ can be precomputed after quantization and stored before ANN-to-SNN conversion, incurring no runtime overhead.

Finally, the residual potential V_{rest} that does not trigger a spike corresponds to a fractional part, denoted by S_{float} , yielding within each quantization group:

$$S_{float} = \frac{V_{rest}}{V_{th}}, \quad (15)$$

$$output_{ANN} = output_{SNN} = S_{int} + S_{float}. \quad (16)$$

This formulation links ANN dequantization to SNN rate-based computation but faces challenges because higher-precision salient values require scaling factors different from normal quantization groups. To ensure equivalence with the ANN, the firing thresholds must also vary accordingly. After quantization, a 1-bit mask code is used to identify salient values. For each group, we pre-compute the normal threshold V_{th}^n and the cross-group salient threshold V_{th}^s . This enables parallel IF operations within groups for normal values and across groups for salient values. Final SNN output for a sequence of activations and weights is given by Eq. 17, and the mix-precision membrane potential process is shown as Fig. 6.

$$Output_{SNN} = (S_{int}^s + S_{float}^s) + \sum_{g=1}^G (S_{int}^n(g) + S_{float}^n(g)). \quad (17)$$

5 Evaluation

We conduct a comprehensive evaluation of SpikeQuant's accuracy and energy efficiency. For accuracy, we report generation perplexity

(a) Perplexity of Llama2 Models						(b) Perplexity of OPT Models					
Methods	Bits	Llama2-7B		Llama2-13B		Methods	Bits	Opt1.3B		Opt2.7B	
		Wikitext2	C4	Wikitext2	C4			Wikitext2	C4	Wikitext2	C4
Fp16	W16A16	5.68	7.08	4.88	6.46	Fp16	W16A16	14.63	14.72	12.47	13.17
SmoothQuant	W4A4	22.62	31.21	33.98	41.53	GroupQuant	W4A4	36.03	31.08	15.19	15.64
OmniQuant	W4A4	14.61	19.35	12.3	15.87	OneBit	W1A16	25.42	22.95	21.86	20.76
Atom	W4A(4&8)	6.16	7.70	5.46	7.03	LRQuant	W2A16	48.43	69.34	30.59	42.49
SpikeLLM	W4A(4&8)	11.36	15.87	9.71	12.10	GPTQ	W3A16	20.97	21.63	16.88	18.17
SpikeQuant	W4A(4&5)	5.79	7.33	5.04	6.65	SpikeQuant	W4A(4&5)	16.38	16.56	13.15	13.91

Table 1: Perplexity (↓) performance of Llama2 and OPT models on Wikitext2 and C4 datasets. The best score is bolded.

and zero-shot accuracy on standard benchmarks. For energy efficiency, we estimate the energy cost of MACs, accumulations, and data movement during inference. We also perform ablation studies to quantify the contribution of each component in SpikeQuant.

5.1 Settings

Training Details. SpikeQuant employs 4-bit asymmetric quantization for both weights and activations without clipping, while salient values are quantized at 5 bits. For calibration, we randomly sample 128 segments of 2048 tokens from WikiText2 and use them to estimate saliency thresholds for each layer and module. Quantization is performed layer by layer, running on two NVIDIA A100 40GB GPUs.

Baselines. To comprehensively assess SpikeQuant’s accuracy performance, we compare against a diverse set of quantization baselines. For Llama2 models, we focus on methods that specifically address saliency in weights and activations to improve quantized accuracy. In particular, SmoothQuant [8] and OmniQuant [30] use 4-bit quantization for both weights and activations (W4A4), while Atom [9] and SpikeLLM [33] adopt W4A4 with salient values elevated to 8 bits. For OPT models, we include baselines spanning different weight quantization bitwidth: GroupQuant, OneBit [34], LRQuant [35], and GPTQ [36]. GroupQuant is a simple group-wise scheme with 128 groups. A complete description of baselines is in the Appendix A.

Evaluation Tasks. We evaluate SpikeQuant on widely used open-source LLaMA [1] and OPT [37] models. For model accuracy performance, we adopt standard metrics, including model accuracy, perplexity, and zero-shot accuracy. Language modeling perplexity is measured on the WikiText-2 [21] and C4 [22] benchmarks. For zero-shot evaluation, we report accuracy on PIQA [38], ARC-easy [39], ARC-challenge [39], BoolQ [40], HellaSwag [39], and Winogrande [41]. For inference energy-cost performance analysis, we use measured energy for compute, data movement, and memory access obtained from a commercial 22 nm process.

5.2 Accuracy Evaluation

As shown in Table 1, we reports perplexity on Llama and OPT. SpikeQuant further lowers perplexity, with 0.11 decrease on WikiText-2 for Llama-7B. On OPT, it largely preserves perplexity, averaging a 0.71 decrease on OPT-2.7B, while existing methods fail to achieve comparable levels.

In Table 2, we compare zero-shot accuracy on six tasks across Llama and OPT models. SpikeQuant consistently outperforms other weight–activation quantization baselines. Among salient aware W4A4 methods, SpikeQuant incurs only 0.09% and 1.45% average accuracy loss on Llama-7B and Llama-13B versus FP16, whereas prior work reports up to 15.81% and 16.76% on Llama-7B and 13B under the same settings. For baselines that reduce weight bit-width to improve compression, SpikeQuant likewise maintains low precision with only 2.60% (OPT-1.3B) and 0.42% (OPT-2.7B) accuracy loss, compared to losses as high as 10.05% and 11.42% for similar methods.

SpikeQuant’s strong accuracy and perplexity stem from precisely identifying salient values and quantizing them in place at higher bit-width. In contrast, Atom and SpikeLLM, although elevating salient values to 8-bit, rely on fixed salient channels or fixed salient ratios, which fail to accurately localize saliency. This can inflate the quantization scale spanning normal activations and salient values, reducing quantization fidelity and ultimately lowering model accuracy. Additional experiments on the number and proportion of salient values are in the Appendix C.1.

5.3 Energy Efficiency Evaluation

Followed by prior works [42, 43], the inference energy in digital circuits comprises computation energy (E_{compute}) and data movement energy (E_{data}):

$$E = E_{\text{compute}} + E_{\text{data}}. \quad (18)$$

The computation energy E_{compute} accounts for arithmetic operations of MACs and ACCs. The data movement energy E_{data} covers the energy to read weights from SRAM and to move data from previous-layer outputs to the current layer’s computation.

Our baselines fall into three categories: typical weight–activation quantization (e.g., SmoothQuant and OneBit), mixed-precision weight–activation quantization (e.g., Atom), and SNN-based mixed-precision weight–activation quantization (e.g., Spike LLM). Our energy model assumes a spatial dataflow architecture with a Network-on-Chip (NoC), a design common to neuromorphic hardware like Loihi [44] and TrueNorth [45] and large-scale AI accelerators such as SambaNova [46] and Cerebras [47]. Based on the dataflow architecture and Eq. (18), we derive energy-cost formulas for these three baseline categories (provided in detail in the Appendix B). To evaluate hardware costs, we synthesized MAC and ACC units at the bit

Table 2: Zero-shot (%) ($\uparrow$) performance of Llama2 and OPT models on six common sense tasks.

Model	Method	PIQA	ARC_e	ARC_c	BoolQ	HellaSwag	Winogrande	Avg.
Opt-1.3B	Fp16	72.42	50.80	29.69	57.68	53.73	59.67	54.00
	GroupQuant	67.79	42.93	25.26	49.51	45.94	54.93	47.73
	OneBit	62.57	41.25	24.06	59.45	34.26	51.14	45.46
	LRQuant	60.23	43.22	19.80	57.98	30.71	51.78	43.95
	GPTQ	68.34	46.17	27.65	51.57	46.53	55.26	49.25
	SpikeQuant	71.49	49.66	27.22	53.27	51.46	58.56	51.94
Opt-2.7B	Fp16	74.81	54.34	31.31	60.28	60.59	60.93	57.04
	GroupQuant	72.91	49.96	29.44	62.26	55.94	59.75	55.04
	OneBit	63.87	43.39	24.40	54.28	38.18	51.67	45.97
	LRQuant	64.04	47.81	20.65	54.10	33.59	53.51	45.62
	GPTQ	71.38	48.19	27.82	55.43	49.86	54.82	51.25
	SpikeQuant	74.21	53.79	31.66	61.19	59.11	59.75	56.62
Llama2-7B	Fp16	77.37	52.53	41.38	73.12	72.99	66.85	64.04
	GroupQuant	75.79	54.04	41.13	71.87	71.68	63.54	63.01
	SmoothQuant	63.11	40.03	31.57	58.47	43.38	52.80	48.23
	OmniQuant	66.15	45.20	31.14	63.51	56.44	53.43	52.65
	Atom	76.28	52.10	38.99	69.79	69.81	63.69	61.78
	SpikeLLM	64.47	48.74	27.30	63.27	43.29	56.83	50.65
	SpikeQuant	77.09	55.39	40.36	73.49	72.09	65.27	63.95
Llama2-13B	Fp16	79.05	59.85	44.62	68.53	76.22	70.09	66.39
	GroupQuant	77.75	55.64	43.69	66.45	74.78	68.3	64.45
	SmoothQuant	64.47	41.75	30.89	62.29	46.68	51.70	49.63
	OmniQuant	69.69	47.39	33.10	62.84	58.96	55.80	54.63
	Atom	77.69	57.58	42.92	67.46	73.77	68.51	64.66
	SpikeLLM	66.49	55.30	30.12	64.16	47.43	51.54	52.51
	SpikeQuant	79.33	56.48	44.03	66.15	75.53	68.11	64.94

Table 3: Energy Ratio of Different Computing Units and Data Movement Operations (Normalized to 4-4-4 ACC)

Process	Computing								Data movement	
	4-4-4 ACC	5-5-5 ACC	4-4-32 MAC	4-5-32 MAC	4-8-32 MAC	1-16-32 MAC	2-16-32 MAC	3-16-32 MAC	Read/bit	Move/bit
Energy	1.00	1.18	8.66	9.24	10.94	10.89	11.46	13.28	6.04	11.04

widths used by each baseline in a commercial 22 nm process. Table 3 reports the resulting energy measurements.

Using the measurements in Table 3 and the energy formulas, we obtain the final energy-cost comparison results shown in Figure 7. The results show that SpikeQuant, while achieving higher accuracy, reduces energy by up to 78.5% (OPT-2.7B, compared to GPTQ). The savings primarily stem from SpikeQuant’s lower quantization bit widths and the fact that, for each activation parameter, only a 1-bit TTFS spike encoding needs to be moved.

5.4 Ablation Study

Table 4 presents the perplexity results of our ablation study on various components in SpikeQuant. As shown, the baseline FP16 models achieve low perplexity across all scales. However, direct

W4A4 quantization causes severe performance degradation, with perplexity increasing by several orders of magnitude, indicating that naive low-bit quantization leads to substantial information loss. Introducing group-wise quantization (group size = 256) mitigates this issue, reducing perplexity to a reasonable level. Further decreasing the group size to 128 yields additional improvements, demonstrating the benefit of finer granularity. When we fix the top 128 salient channels, the perplexity does not improve, suggesting that salient activations vary dynamically across forward passes and cannot be captured by static channel selection.

In contrast, SpikeQuant adaptively identifies salient channels at runtime using a MAD-based criterion, allowing flexible precision allocation without introducing instability. This adaptive mechanism enables SpikeQuant to achieve significantly lower perplexity than

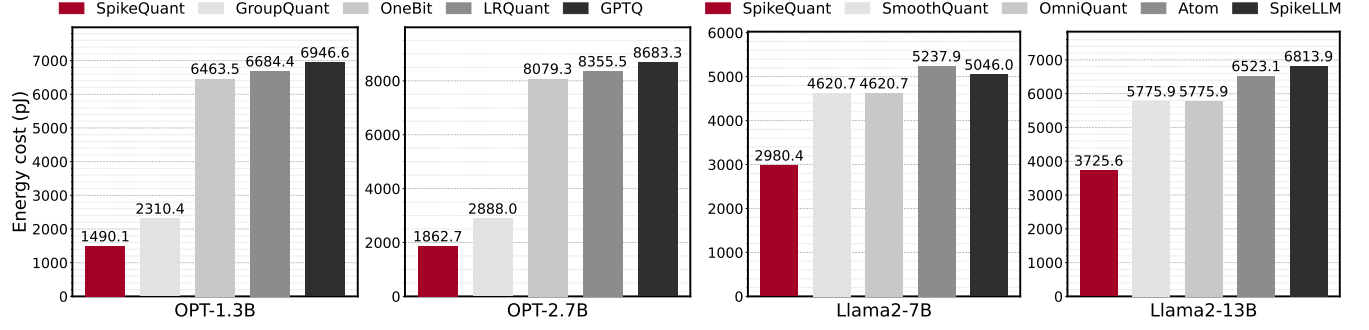


Figure 7: Energy cost (J) comparison across OPT and Llama models

Table 4: Perplexity (J) ablation study on different quantization techniques used in SpikeQuant

Method	OPT-1.3B	OPT-2.7B	Llama2-7B	Llama2-13B
FP16	14.68	12.82	6.38	5.67
+ W4A4	33402.11 (+33387.43)	12668.00 (+12655.18)	660.33 (+653.95)	411.64 (+405.97)
+ Group size 256	259.10 (-33143.01)	83.12 (-12584.88)	7.16 (-653.17)	6.29 (-405.35)
Group size 128	33.55 (-225.55)	15.41 (-67.71)	6.84 (-0.32)	6.02 (-0.27)
+ Fix 128 salient channel	35.10 (+1.55)	17.28 (+1.87)	6.93 (+0.09)	6.25 (+0.23)
+ SpikeQuant (MAD $r = 5$)	16.54 (-18.56)	13.55 (-3.73)	6.60 (-0.33)	5.86 (-0.39)
SpikeQuant (MAD $r = 3.5$)	16.47 (-0.07)	13.53 (-0.02)	6.56 (-0.04)	5.85 (-0.01)

all static quantization variants. Notably, even when the baseline perplexity is already close to the FP16 level, SpikeQuant continues to yield measurable reductions (e.g., from 35.10 to 16.47 for OPT-1.3B and from 6.25 to 5.85 for Llama2-13B)

6 Related Works

Model quantization. Quantization reduces model size by converting floating-point parameters to low-bit representations, typically down to 2, 4, or 8 bits [48–50], enabling efficient deployment on resource-constrained devices [51, 52]. The progression from weight quantization [53, 54] to weight-activation quantization [20, 55] has significantly advanced neural network compression. However, prior works observed that a key challenge in LLM quantization is the salient values (also called outliers) phenomenon [7, 10]. These large-magnitude values are challenging as they distort the quantization range, leading to significant precision loss for other values [56]. Most existing methods mitigate this by focusing on fixed salient channels in weights or activations [9, 57, 58]. This approach, however, overlooks that activation distributions are dynamic and vary with each forward pass during inference.

Spiking neural networks. Spiking Neural Networks (SNNs) [59, 60] emerge as a promising brain-inspired computational model that mimics the binary spike information transmission mechanism of biological neurons, drawing from fundamental neuroscience principles of neural communication through action potentials [61, 62]. Current SNNs research predominantly focuses on computer vision tasks [13, 63] and small-scale language processing tasks [64, 65] by converting ANNs to SNNs, where SNNs offer energy-efficient alternatives to traditional ANNs due to their high biological plausibility.

These ANN-to-SNN methods include quantization of weights and membrane dynamics for efficiency [13], ANN parameter initialization followed by layer-wise distillation to bridge heterogeneous representations [63], and architectural adaptations of pretrained models to spiking formats for language tasks [64, 65]. However, applications of SNNs to LLMs remain significantly underexplored, representing a substantial gap in current research directions despite the potential for energy-efficient language processing that could leverage the temporal dynamics inherent in spiking mechanisms.

7 Conclusion

The rapid rise of ANN-based LLMs creates deployment demands in Web-of-Things (WoT) settings with strict privacy, latency, and energy constraints. We propose SpikeQuant, a brain-inspired quantization framework that embeds spiking dynamics into LLM computation. By encoding salient activations as spike counts and mapping quantization scales to firing thresholds, SpikeQuant enables dequantization-free, energy-efficient inference. Experiments show near-FP16 accuracy under W4A4 and even reduced perplexity when baselines are strong, highlighting its promise for low-power, high-accuracy LLMs in WoT.

Acknowledgment

This research was supported by the National Natural Science Foundation of China (No. 62172454, No. U25B2002), the Guangdong Basic and Applied Basic Research Foundation (No. 2023B1515120058, No. 2023B1515020120), the Guangzhou Basic and Applied Basic Research Scheme (No. 2024A04J3517), and the Ministry of Education, Singapore, under the Academic Research Fund Tier 1 (FY2024).

References

- [1] Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmin Babaei, Nikolay Bashlykov, Soumya Batra, Prajwal Bhargava, Shrutu Bhosale, et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023.
- [2] Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, L  lio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timoth  e Lacroix, and William El Sayed. Mistral 7b. *arXiv preprint arXiv:2310.06825*, 2023.
- [3] Charlie F Ruan, Yucheng Qin, Xun Zhou, Ruihang Lai, Hongyi Jin, Yixin Dong, Bohan Hou, Meng-Shiun Yu, Yiyan Zhai, Sudeep Agarwal, et al. Webllm: A high-performance in-browser llm inference engine. *arXiv preprint arXiv:2412.15803*, 2024.
- [4] Xiao Liu, Hanyu Lai, Hao Yu, Yifan Xu, Aohan Zeng, Zhengxiao Du, Peng Zhang, Yuxiao Dong, and Jie Tang. Webglm: towards an efficient web-enhanced question answering system with human preferences. In *Proceedings of the 29th ACM SIGKDD conference on knowledge discovery and data mining*, pages 4549–4560, 2023.
- [5] Soham Poddar, Paramita Koley, Janardan Misra, Niloy Ganguly, and Saptarshi Ghosh. Towards sustainable NLP: Insights from benchmarking inference energy in large language models. In Luis Chiruzzo, Alan Ritter, and Lu Wang, editors, *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 12688–12704, Albuquerque, New Mexico, April 2025. Association for Computational Linguistics.
- [6] Luchang Li, Sheng Qian, Jie Lu, Lunxi Yuan, Rui Wang, and Qin Xie. Transformer-lite: High-efficiency deployment of large language models on mobile phone gpus, 2024.
- [7] Yongqi An, Xu Zhao, Tao Yu, Ming Tang, and Jinqiao Wang. Systematic outliers in large language models. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [8] Guangxuan Xiao, Ji Lin, Mickael Seznec, Hao Wu, Julien Demouth, and Song Han. Smoothquant: Accurate and efficient post-training quantization for large language models. In *International conference on machine learning*, pages 38087–38099. PMLR, 2023.
- [9] Yilong Zhao, Chien-Yu Lin, Kan Zhu, Zihao Ye, Lequn Chen, Size Zheng, Luis Ceze, Arvind Krishnamurthy, Tianqi Chen, and Baris Kasikci. Atom: Low-bit quantization for efficient and accurate llm serving. In P. Gibbons, G. Pekhimenko, and C. De Sa, editors, *Proceedings of Machine Learning and Systems*, volume 6, pages 196–209, 2024.
- [10] Ji Lin, Jiaming Tang, Haotian Tang, Shang Yang, Guangxuan Xiao, and Song Han. Awq: Activation-aware weight quantization for on-device llm compression and acceleration. *GetMobile: Mobile Comp. and Comm.*, 28(4):12–17, January 2025.
- [11] Catherine D Schuman, Shrutu R Kulkarni, Maryam Parsa, J Parker Mitchell, Prasanna Date, and Bill Kay. Opportunities for neuromorphic computing algorithms and applications. *Nature Computational Science*, 2(1):10–19, 2022.
- [12] Yufei Guo, Xuhui Huang, and Zhe Ma. Direct learning-based deep spiking neural networks: a review. *Frontiers in Neuroscience*, 17:1209795, 2023.
- [13] Wenjie Wei, Yu Liang, Ammar Belatreche, Yichen Xiao, Honglin Cao, Zhenbang Ren, Guoqing Wang, Malu Zhang, and Yang Yang. Q-snns: Quantized spiking neural networks. In *Proceedings of the 32nd ACM international conference on multimedia*, pages 8441–8450, 2024.
- [14] Wenjie Wei, Malu Zhang, Zijian Zhou, Ammar Belatreche, Yimeng Shan, Yu Liang, Honglin Cao, Jieyuan Zhang, and Yang Yang. QP-SNN: Quantized and pruned spiking neural networks. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [15] Haomin Li, Fangxin Liu, Zewen Sun, Zongwu Wang, Shiyuan Huang, Ning Yang, and Li Jiang. *NeuronQuant: Accurate and Efficient Post-Training Quantization for Spiking Neural Networks*, page 734–740. Association for Computing Machinery, New York, NY, USA, 2025.
- [16] Malyaban Bal, Yi Jiang, and Abhronil Sengupta. Exploring extreme quantization in spiking language models. In *2024 International Conference on Neuromorphic Systems (ICONS)*, pages 272–276, 2024.
- [17] Lina Bonilla, Jacques Gautrais, Simon Thorpe, and Timoth  e Masquelier. Analyzing time-to-first-spike coding schemes: A theoretical approach. *Frontiers in Neuroscience*, 16:971937, 2022.
- [18] Youngeun Kim, Hyoungseob Park, Abhishek Moitra, Abhiroop Bhattacharjee, Yeshwanth Venkatesha, and Priyadarshini Panda. Rate coding or direct coding: Which one is better for accurate, robust, and energy-efficient spiking neural networks? In *ICASSP 2022 - 2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 71–75, 2022.
- [19] Wenzhe Guo, Mohammed E Fouda, Ahmed M Eltwail, and Khaled Nabil Salama. Neural coding in spiking neural networks: A comparative study for robust neuromorphic systems. *Frontiers in Neuroscience*, 15:638474, 2021.
- [20] Janghwan Lee, Minsoo Kim, Seungcheol Baek, Seok Joong Hwang, Wonyong Sung, and Jungwook Choi. Enhancing computation efficiency in large language models through weight and activation quantization. *arXiv preprint arXiv:2311.05161*, 2023.
- [21] Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. Pointer sentinel mixture models. In *International Conference on Learning Representations*, 2017.
- [22] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of machine learning research*, 21(140):1–67, 2020.
- [23] Yelysei Bondarenko, Markus Nagel, and Tijmen Blankevoort. Quantizable transformers: removing outliers by helping attention heads do nothing. In *Proceedings of the 37th International Conference on Neural Information Processing Systems*, NIPS ’23, Red Hook, NY, USA, 2023. Curran Associates Inc.
- [24] Patrik Cz    , G    r K        , and S      r Sz        . Addressing activation outliers in llms: A systematic review of post-training quantization techniques. *IEEE Access*, 2025.
- [25] Tim Dettmers, Mike Lewis, Younes Belkada, and Luke Zettlemoyer. Gpt3.int8 (): 8-bit matrix multiplication for transformers at scale. *Advances in neural information processing systems*, 35:30318–30332, 2022.
- [26] Amir Ardakani, Arash Ardakani, Brett Meyer, James J Clark, and Warren J Gross. Standard deviation-based quantization for deep neural networks. *arXiv preprint arXiv:2202.12422*, 2022.
- [27] Eunhyeok Park, Sungjoo Yoo, and Peter Vajda. Value-aware quantization for training and inference of neural networks. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 580–595, 2018.
- [28] Peter J Rousseeuw and Christophe Croux. Alternatives to the median absolute deviation. *Journal of the American Statistical association*, 88(424):1273–1283, 1993.
- [29] Boris Iglewicz and David C Hoaglin. *Volume 16: how to detect and handle outliers*. Quality Press, 1993.
- [30] Wenqi Shao, Mengzhao Chen, Zhaoyang Zhang, Peng Xu, Lirui Zhao, Zhiqian Li, Kaipeng Zhang, Peng Gao, Yu Qiao, and Ping Luo. Omniquant: Omnidirectionally calibrated quantization for large language models. In *The Twelfth International Conference on Learning Representations*, 2024.
- [31] Jiaming Yang, Chenwei Tang, Caiyang Yu, and Jiancheng Lv. Gwq: Group-wise quantization framework for neural networks. In *Asian Conference on Machine Learning*, pages 1526–1541. PMLR, 2024.
- [32] Mariam Rakka, Rachid Karami, Ahmed M Eltwail, Mohammed E Fouda, and Fadi Kurdahi. Bf-imna: A bit fluid in-memory neural architecture for neural network acceleration. *arXiv preprint arXiv:2411.01417*, 2024.
- [33] Xingrun Xing, Boyan Gao, Zheng Liu, David A. Clifton, Shitao Xiao, Wanpeng Zhang, Li Du, Zheng Zhang, Guoqi Li, and Jiajun Zhang. SpikeLLM: Scaling up spiking neural network to large language models via saliency-based spiking. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [34] Yuzhuang Xu, Xu Han, Zonghan Yang, Shuo Wang, Qingfu Zhu, Zhiyuan Liu, Weidong Liu, and Wanxiang Che. Onebit: Towards extremely low-bit large language models. *Advances in Neural Information Processing Systems*, 37:66357–66382, 2024.
- [35] Jiaqi Zhao, Miao Zhang, Chao Zeng, Ming Wang, Xuebo Liu, and Liqiang Nie. Lrquant: Learnable and robust post-training quantization for large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2240–2255, 2024.
- [36] Elias Frantar, Saleh Ashkboos, Torsten Hoefler, and Dan Alistarh. Gptq: Accurate post-training quantization for generative pre-trained transformers. *arXiv preprint arXiv:2210.17323*, 2022.
- [37] Susan Zhang, Stephen Roller, Naman Goyal, Mikel Artetxe, Moya Chen, Shuohui Chen, Christopher Dewan, Mona Diab, Xian Li, Xi Victoria Lin, et al. Opt: Open pre-trained transformer language models. *arXiv preprint arXiv:2205.01068*, 2022.
- [38] Yonatan Bisk, Rowan Zellers, Jianfeng Gao, Yejin Choi, et al. Piqa: Reasoning about physical commonsense in natural language. In *Proceedings of the AAAI conference on artificial intelligence*, volume 34, pages 7432–7439, 2020.
- [39] Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. Think you have solved question answering? try arc, the ai2 reasoning challenge. *arXiv preprint arXiv:1803.05457*, 2018.
- [40] Christopher Clark, Kenton Lee, Ming-Wei Chang, Tom Kwiatkowski, Michael Collins, and Kristina Toutanova. Boolq: Exploring the surprising difficulty of natural yes/no questions. *arXiv preprint arXiv:1905.10044*, 2019.
- [41] Keisuke Sakaguchi, Ronan Le Bras, Chandra Bhagavatula, and Yejin Choi. Winogrande: An adversarial winograd schema challenge at scale. *Communications of the ACM*, 64(9):99–106, 2021.
- [42] Zhanglu Yan, Zhenyu Bai, and Weng-Fai Wong. Reconsidering the energy efficiency of spiking neural networks. *arXiv preprint arXiv:2409.08290*, 2024.
- [43] Zhanglu Yan, Jiayi Mao, Qianhui Liu, Fanfan Li, Gang Pan, Tao Luo, Bowen Zhu, and Weng-Fai Wong. Otters: An energy-efficient spikingtransformer via optical time-to-first-spike encoding, 2025.
- [44] Andrew Lines, Prasad Joshi, Ruokun Liu, Steve McCoy, Jonathan Tse, Yi-Hsin Weng, and Mike Davies. Loihi asynchronous neuromorphic research chip. In

- 2018 24th IEEE International Symposium on Asynchronous Circuits and Systems (ASYNC), pages 32–33, 2018.
- [45] Filipp Akopyan, Jun Sawada, Andrew Cassidy, Rodrigo Alvarez-Icaza, John Arthur, Paul Merolla, Nabil Imam, Yutaka Nakamura, Pallab Datta, Gi-Joon Nam, et al. Truenorth: Design and tool flow of a 65 mw 1 million neuron programmable neurosynaptic chip. *IEEE transactions on computer-aided design of integrated circuits and systems*, 34(10):1537–1557, 2015.
- [46] Raghu Prabhakar, Sumti Jairath, and Jinuk Luke Shin. Sambanova sn10 rdu: A 7nm dataflow architecture to accelerate software 2.0. In *2022 IEEE International Solid-State Circuits Conference (ISSCC)*, volume 65, pages 350–352. IEEE, 2022.
- [47] Sean Lie. Cerebras architecture deep dive: First look inside the hw/sw co-design for deep learning: Cerebras systems. In *2022 IEEE Hot Chips 34 Symposium (HCS)*, pages 1–34. IEEE Computer Society, 2022.
- [48] Sayeed Shafayet Chowdhury, Isha Garg, and Kaushik Roy. Spatio-temporal pruning and quantization for low-latency spiking neural networks. In *2021 International Joint Conference on Neural Networks (IJCNN)*, pages 1–9. IEEE, 2021.
- [49] Lei Deng, Yujie Wu, Yifan Hu, Ling Liang, Guoqi Li, Xing Hu, Yufei Ding, Peng Li, and Yuan Xie. Comprehensive snn compression using admm optimization and activity regularization. *IEEE transactions on neural networks and learning systems*, 34(6):2791–2805, 2021.
- [50] Martino Sorbaro, Qian Liu, Massimo Bortone, and Sadique Sheik. Optimizing the energy consumption of spiking neural networks for neuromorphic applications. *Frontiers in neuroscience*, 14:662, 2020.
- [51] Amir Gholami, Sehoon Kim, Zhen Dong, Zhewei Yao, Michael W Mahoney, and Kurt Keutzer. A survey of quantization methods for efficient neural network inference. In *Low-power computer vision*, pages 291–326. Chapman and Hall/CRC, 2022.
- [52] Tailin Liang, John Glossner, Lei Wang, Shaobo Shi, and Xiaotong Zhang. Pruning and quantization for deep neural network acceleration: A survey. *Neurocomputing*, 461:370–403, 2021.
- [53] Shuhei Kashiwamura, Ayaka Sakata, and Masaaki Imaizumi. Effect of weight quantization on learning models by typical case analysis. In *2024 IEEE International Symposium on Information Theory (ISIT)*, pages 357–362. IEEE, 2024.
- [54] Pranav Ajit Nair and Arun Sai Suggala. Cdquant: Greedy coordinate descent for accurate llm quantization, 2024.
- [55] Yundong Gai and Ping Li. Atq: Activation transformation for weight-activation quantization of large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 17187–17194, 2024.
- [56] Haokun Lin, Haobo Xu, Yichen Wu, Jingzhi Cui, Yingtao Zhang, Linzhan Mou, Linqi Song, Zhenan Sun, and Ying Wei. Duquant: Distributing outliers via dual transformation makes stronger quantized llms, 2024.
- [57] Xiuying Wei, Yunchen Zhang, Yuhang Li, Xiangguo Zhang, Ruihao Gong, Jinyang Guo, and Xianglong Liu. Outlier suppression+: Accurate quantization of large language models by equivalent and effective shifting and scaling. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 1648–1665, Singapore, December 2023. Association for Computational Linguistics.
- [58] Aniruddha Nrusimha, Mayank Mishra, Naigang Wang, Dan Alistarh, Rameswar Panda, and Yoon Kim. Mitigating the impact of outlier channels for language model quantization with activation regularization. *arXiv preprint arXiv:2404.03605*, 2024.
- [59] Kaushik Roy, Akhilesh Jaiswal, and Priyadarshini Panda. Towards spike-based machine intelligence with neuromorphic computing. *Nature*, 575(7784):607–617, 2019.
- [60] Wolfgang Maass. Networks of spiking neurons: the third generation of neural network models. *Neural networks*, 10(9):1659–1671, 1997.
- [61] Adnan Mehonic and Anthony J Kenyon. Brain-inspired computing needs a master plan. *Nature*, 604(7905):255–260, 2022.
- [62] Youhui Zhang, Peng Qu, Yu Ji, Weihao Zhang, Guangrong Gao, Guanrui Wang, Sen Song, Guoqi Li, Wenguang Chen, Weimin Zheng, et al. A system hierarchy for brain-inspired computing. *Nature*, 586(7829):378–384, 2020.
- [63] Di Hong, Yu Qi, and Yueming Wang. Lasnn: Layer-wise ann-to-snn distillation for effective and efficient training in deep spiking neural networks. *Neurocomputing*, page 131351, 2025.
- [64] Malyaban Bal and Abhronil Sengupta. Spikingbert: Distilling bert to train spiking language models using implicit differentiation. In *Proceedings of the AAAI conference on artificial intelligence*, volume 38, pages 10998–11006, 2024.
- [65] Rui-Jie Zhu, Qihang Zhao, Guoqi Li, and Jason K Eshraghian. Spikegpt: Generative pre-trained language model with spiking neural networks. *arXiv preprint arXiv:2302.13939*, 2023.

A Baseline details

SmoothQuant [8] is a post-training quantization (PTQ) method that transfers part of the quantization “difficulty” from activations to weights. It rescales activation channels and applies corresponding compensation to the weights, making activations easier to represent at low bitwidth. This approach enables W8A8 (8-bit weight and activation) quantization while maintaining minimal accuracy loss on large language models.

OmniQuant [30] is a differentiable and learnable quantization framework featuring two key modules: Learnable Weight Clipping, which regulates extreme weight values, and Learnable Equivalent Transformation, which shifts activation outliers into the weight domain. By minimizing quantization error at the block level, OmniQuant achieves strong performance even under extremely low-bit settings such as W4A4 or W3A16.

Atom [9] targets service-oriented, throughput-optimized low-bit deployment. It combines mixed-precision, fine-grained group quantization, dynamic activation quantization, and KV-cache quantization, enabling efficient 4-bit weight and activation inference. Atom achieves substantial speedups over FP16 and INT8 while maintaining competitive accuracy.

SpikeLLM [33] integrates Spiking Neural Network (SNN) principles with quantization and discretization mechanisms for language modeling. It introduces an “Optimal Brain Spiking” quantization framework that selectively applies spiking or quantization to specific channels, reducing energy consumption and accelerating inference. Moreover, SpikeLLM is compatible with existing quantization pipelines such as GPTQ and OmniQuant.

OneBit [34] explores extreme low-bit quantization, compressing LLM weight matrices to 1 bit (or near-binary precision). It designs specialized 1-bit parameter representations and matrix-factorization-based initialization to improve convergence after quantization. Experiments show that OneBit can achieve reasonable performance retention on models such as LLaMa under ultra-low bitwidths.

LRQuant [35] is a learnable and robust PTQ approach built upon SmoothQuant and OmniQuant. It introduces a new NLC + MSE loss and a learnable smoothing-parameter mechanism. Additionally, it proposes Gamma Migration, which shifts the LayerNorm γ parameter into subsequent layer weights or residual paths, thereby stabilizing activation outlier distributions and improving quantization robustness.

GPTQ [36] is a one-shot weight quantization method tailored for GPT- and OPT-style generative transformers. It leverages approximate second-order statistics (e.g., Hessian or inverse-variance approximations) to guide quantization, maintaining low error even at 3–4 bit precision. OPTQ enables large models to be quantized within a few GPU hours with negligible accuracy degradation.

B Energy Cost Formulation

We model the energy cost of a linear transformation within one Transformer block under different quantization and encoding schemes. The following notations are used throughout:

- B , S , H , and Out denote the batch size, sequence length, hidden dimension, and output dimension, respectively.
- b_w and b_a represent the bitwidths of weights and activations.

Table 5: Ratio and number of salient values comparison in OPT and Llama2 models

Methods	OPT1.3B (H=2048)		OPT2.7B (H=2560)		Llama2-7B (H=4096)		Llama2-13B (H=5120)	
	Ratio	Number	Ratio	Number	Ratio	Number	Ratio	Number
SpikeQuant	2.45%	50	2.15%	55	2.68%	110	2.30%	118
Atom	6.25%	128	5.00%	128	3.12%	128	2.50%	128
SpikeLLM	10.00%	205	10.00%	256	10.00%	410	10.00%	512

- E_{MAC} and E_{ACC} are the per-operation energy costs of a multiply-accumulate (MAC) and accumulate (ACC) operation.
- E_{read} and E_{move} denote the energy costs of memory read and data movement.
- γ indicates the number of salient activation channels within a hidden group.
- T^{high} and T^{low} are the spike time window for salient and non-salient activations, and S_r^{high} , S_r^{low} represent their spike rates.

For typical quantization, the computation term scales with the total number of MAC operations, while the data movement term reflects memory access proportional to weight and activation bitwidths. The energy cost of a linear transform in one block is:

$$E_q = \underbrace{B \cdot S \cdot Out \cdot H \cdot E_{MAC}}_{E_{compute}} + \underbrace{In \cdot Out \cdot b_w \cdot E_{read} + B \cdot S \cdot H \cdot b_a \cdot E_{move}}_{E_{data}} \quad (19)$$

For mix-precision quantization with γ salient activation values in a hidden group, we partition the energy into high-bit and low-bit computation paths. The high-bit path handles salient activations, consuming more MAC energy and data bandwidth, while the low-bit path handles the rest. The total energy of a linear transform in one block is their sum:

$$E_m = E_q^{high} + E_q^{low} \quad (20)$$

$$E_q^{high} = \underbrace{B \cdot S \cdot Out \cdot \gamma \cdot E_{MAC}^{high}}_{E_{compute}} + \underbrace{\gamma \cdot Out \cdot b_w \cdot E_{read} + B \cdot S \cdot \gamma \cdot b_a^{high} \cdot E_{move}}_{E_{data}} \quad (21)$$

$$E_q^{low} = \underbrace{B \cdot S \cdot Out \cdot (H - \gamma) \cdot E_{MAC}^{low}}_{E_{compute}} + \underbrace{(H - \gamma) \cdot Out \cdot b_w \cdot E_{read} + B \cdot S \cdot (H - \gamma) \cdot b_a^{low} \cdot E_{move}}_{E_{data}} \quad (22)$$

For TTFS encoding based SpikeQuant with γ salient activation values in a hidden group, the energy cost of a linear transform in one block is

$$E_s = E_s^{high} + E_s^{low} \quad (23)$$

$$E_s^{high} = \underbrace{B \cdot S \cdot Out \cdot \gamma \cdot T^{high} \cdot S_r^{high} \cdot (E_{ACC}^{high} + E_{MAC}^{high})}_{E_{compute}} + \underbrace{\gamma \cdot Out \cdot b_w \cdot E_{read} + B \cdot S \cdot \gamma \cdot T^{high} \cdot S_r^{high} \cdot E_{move}}_{E_{data}} \quad (24)$$

$$E_s^{low} = \underbrace{B \cdot S \cdot Out \cdot (H - \gamma) \cdot T^{low} \cdot S_r^{low} \cdot (E_{ACC}^{low} + E_{MAC}^{low})}_{E_{compute}} + \underbrace{(H - \gamma) \cdot Out \cdot b_w \cdot E_{read} + B \cdot S \cdot (H - \gamma) \cdot T^{low} \cdot S_r^{low} \cdot E_{move}}_{E_{data}} \quad (25)$$

C Additional Experiments Results

C.1 Salient Value Ratio and Counts

Table 5 compares the ratio and number of salient activation values identified by different methods with varying hidden dimensions. As shown, SpikeQuant consistently detects a much smaller proportion of salient activations (around 2–3%) compared to Atom (2.5–6.25%) and SpikeLLM (10%). This indicates that SpikeQuant achieves more precise and selective saliency detection, effectively distinguishing truly critical activation channels while maintaining the high accuracy.

C.2 Activations Distribution

Figure 8 illustrates the activation distribution of Llama2-7B during inference on the token “is”. The red lines indicate salient activations identified by the MAD-based algorithm, while the gray ones represent normal activations. From the results, we observe that even within a single forward pass for the same token, the distribution and number of salient values vary across layers and modules. This further demonstrates the rationality and flexibility of SpikeQuant in dynamically determining both the quantity and locations of salient values.

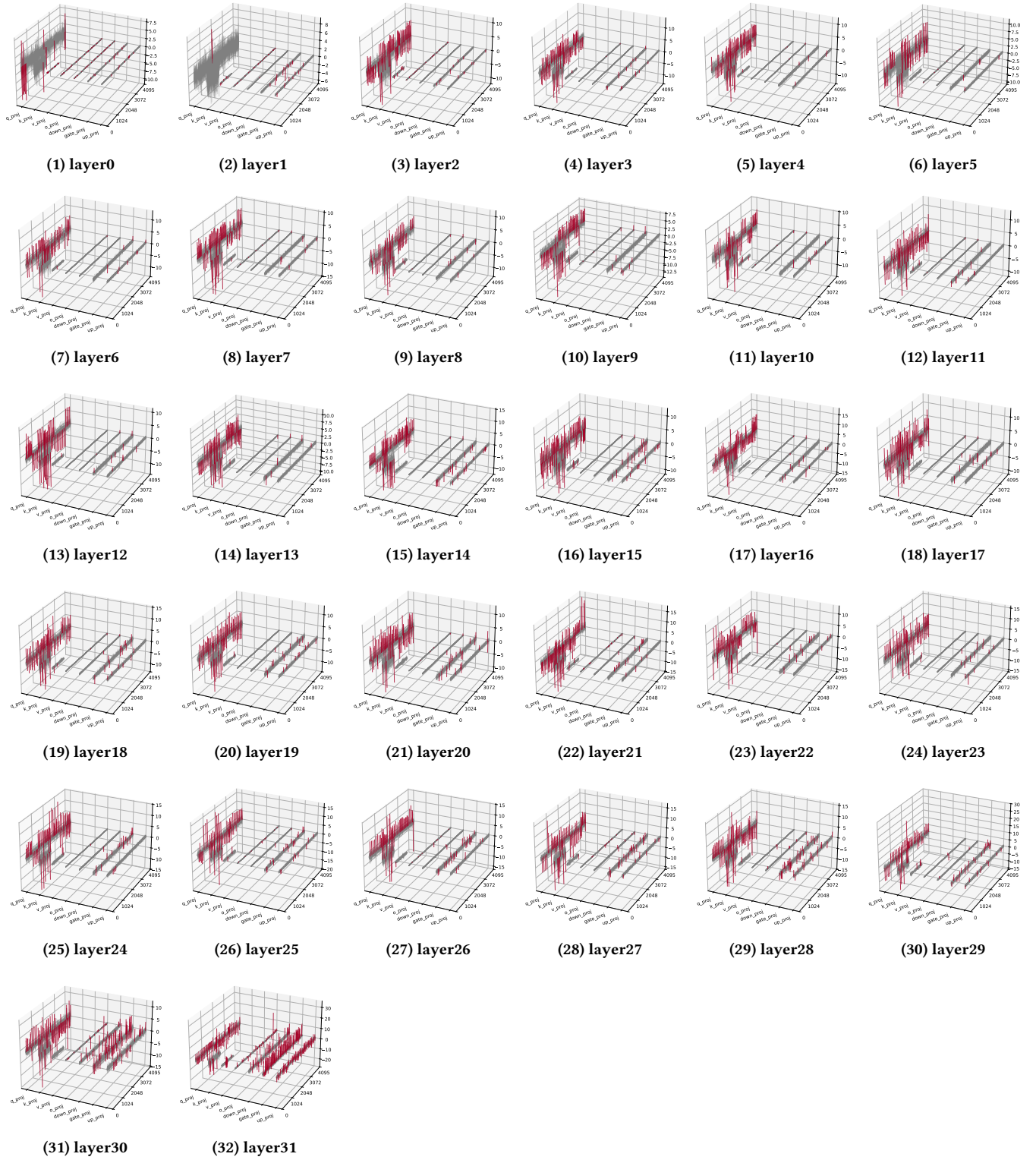


Figure 8: Activation distribution and salient values distribution on Llama2-7B